PARENT FUNCTION

Profile of the cube root

$f(x) = \sqrt[3]{x}$

ℝ DOMAIN · 1-TO-1 INVERTIBLE



KEY POINTS

1 THROUGH
$(0, 0)$

2 PERFECT CUBES
$(\pm 1, \pm 1)$ $(\pm 8, \pm 2)$

GLOBAL

DOMAIN
$(-\infty, \infty)$
every real number has a real cube root

RANGE
$(-\infty, \infty)$

SIGN
preserved
negative in, negative out

ONE-TO-ONE
yes
inverse: the cubing function